INCH-POUND

MIL-M-28787/214(SH)
AMENDMENT 3
6 January 1992
SUPERSEDING
AMENDMENT 2
20 September 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, SERIAL/PARALLEL, DIGITAL, KEY CODE EQF

This amendment forms a part of Military Specification MIL-M-28787/214(EC), dated 17 April 1978, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

PAGE 1

In bottom right hand corner delete "FSC MISC" and substitute "FSC 5963".

PAGE 4

3.5 8, second line: Delete "M28787/214" and substitute "M28787/214-1".

PAGE 6

* TABLE II, Test conditions D1-D33, Test type column: Delete "X" and substitute "S".

* TABLE II, Test conditions E1-E33, Test type column: Delete "X" and substitute "S".

* TABLE II, Test conditions G1-G5, Test type column: Delete "X" and substitute "S"

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PAGE 7

* TABLE II, Test conditions N1-N33, N34-N66, P1-P33 and P34-P66. Delete these test conditions in their entirety.

PAGES 8 AND 9

* TABLE II, Test conditions T1, T3, T5, and T7, Min columns· Delete "130" (16 places) and substitute "115" (16 places)

* TABLE II, Test conditions T1, T3, T5, and T7, Max columns Delete "150" (16 places) and substitute "165" (16 places).

* TABLE II, Test conditions T2, T4, T6, and T8. Min columns Delete "60" (16 places) and substitute "45" (16 places)

* TABLE II, Test conditions T2, T4, T6, and T8, Max columns Delete "80" (16 places) and substitute "95" (16 places)

PAGE 10

TABLE II, Test conditions U1-U66, Max columns Delete "800" (4 places) and substitute "1200" (4 places).

TABLE II, Test conditions Y3-Y5, Max columns Delete "90" (4 places) and substitute "100" (4 places).

PAGES 11 AND 12

4.3. Delete and substitute the following:

"4.3 Test equipment Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

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PAGE 12

* 4.4.2: Delete and substitute the following:

"4 4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions Test conditions A, B, C, D, E, and G are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, E, or G are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, and G where applicable, followed by the remaining tests in the sequence."

* 4.5.1: Delete the third sentence and substitute the following.

"When electrical testing is called for at the end of each temperature stabilization period, test conditions F and H through X shall be performed in the order specified in 4 4.2 using the initial limits for applicable operating temperature "

PAGE 13

* 4.5.3. Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests F, and H through X shall be performed in the order specified in 4 4.2 using the operating temperature end-of-life requirements in table II Following the return of the module to 25 ± 5°C, electrical tests F, and H through Y shall be performed in the order specified in 4 4.2 using the 25 ± 5°C ambient end-of-life requirements in table II."

Add the following new paragraph:

"6.4 Cross reference of classification. Part number M28787/214-1 supersedes part number M28787/214. The two parts are interchangeable."

PAGE 22

FIGURE 7. Delete and substitute new figure 7 as printed on page 7 of this amendment

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PAGE 50

TABLE VIII, NOTE 2, second line: Delete "V03" and substitute "M02".

PAGE 51

FIGURE 14: Delete and substitute new figure 14 as printed on page 8 of this amendment

PAGE 52

FIGURE 15, NOTE 2, second and sixth lines. Delete "V03" (two places) and substitute "M02" (two places).

PAGE 54

FIGURE 16 Delete and substitute new figure 16 as printed on page 9 of this amendment.

PAGE 59

FIGURE 17 Delete and substitute new figure 17 as printed on page 10 of this amendment.

PAGES 68, 69, 70 AND 71

* TABLE XV, Delete in its entirety.

PAGES 72, 73, 74 AND 75

* TABLE XVI, Delete in its entirety.

PAGES 94 AND 95

* TABLE XX, Test conditions T1, T3, T5, and T7, Min columns: Delete "130" (4 places) and substitute "115" (4 places)

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* TABLE XX, Test conditions T1, T3, T5, and T7, Max columns: Delete "150" (4 places) and substitute "165" (4 places).

* TABLE XX, Test conditions T2, T4, T6, and T8, Min columns: Delete "60" (4 places) and substitute "45" (4 places).

* TABLE XX, Test conditions T2, T4, T6, and T8, Max columns Delete "80" (4 places) and substitute "95" (4 places).

PAGE 102

* FIGURE 30: Delete and substitute new figure 30 as printed on page 11 of this amendment

PAGES 110 THROUGH 118

TABLE XXI, Test conditions U1 through U66, Max column. Delete "800" (18 places) and substitute "1200" (18 places).

PAGE 132

FIGURE 38: Delete and substitute new figure 38 as printed on page 12 of this amendment.

PAGE 135

FIGURE 40, NOTE 1. Delete "$t_{TLH} = t_{THL} = 4 \pm 2$ns" and substitute "$t_{TLH} = t_{THL} = 10 \pm 2$ns"

PAGE 137

TABLE XXV, Test conditions Y3 through Y5, Max column· Delete "90" (3 places) and substitute "100" (3 places).

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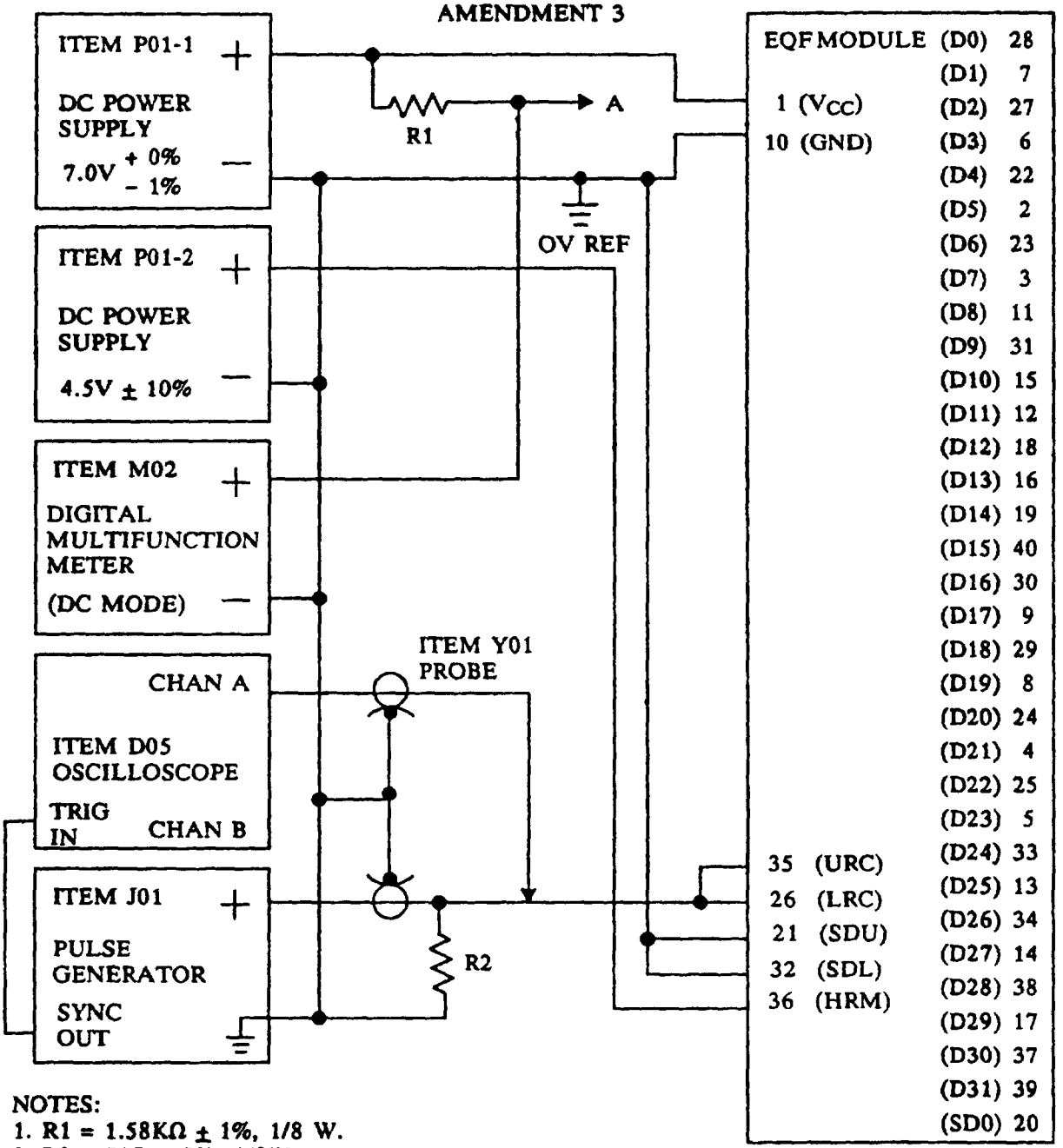
NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - SH

(Project 5963-N779)

Agent: NW

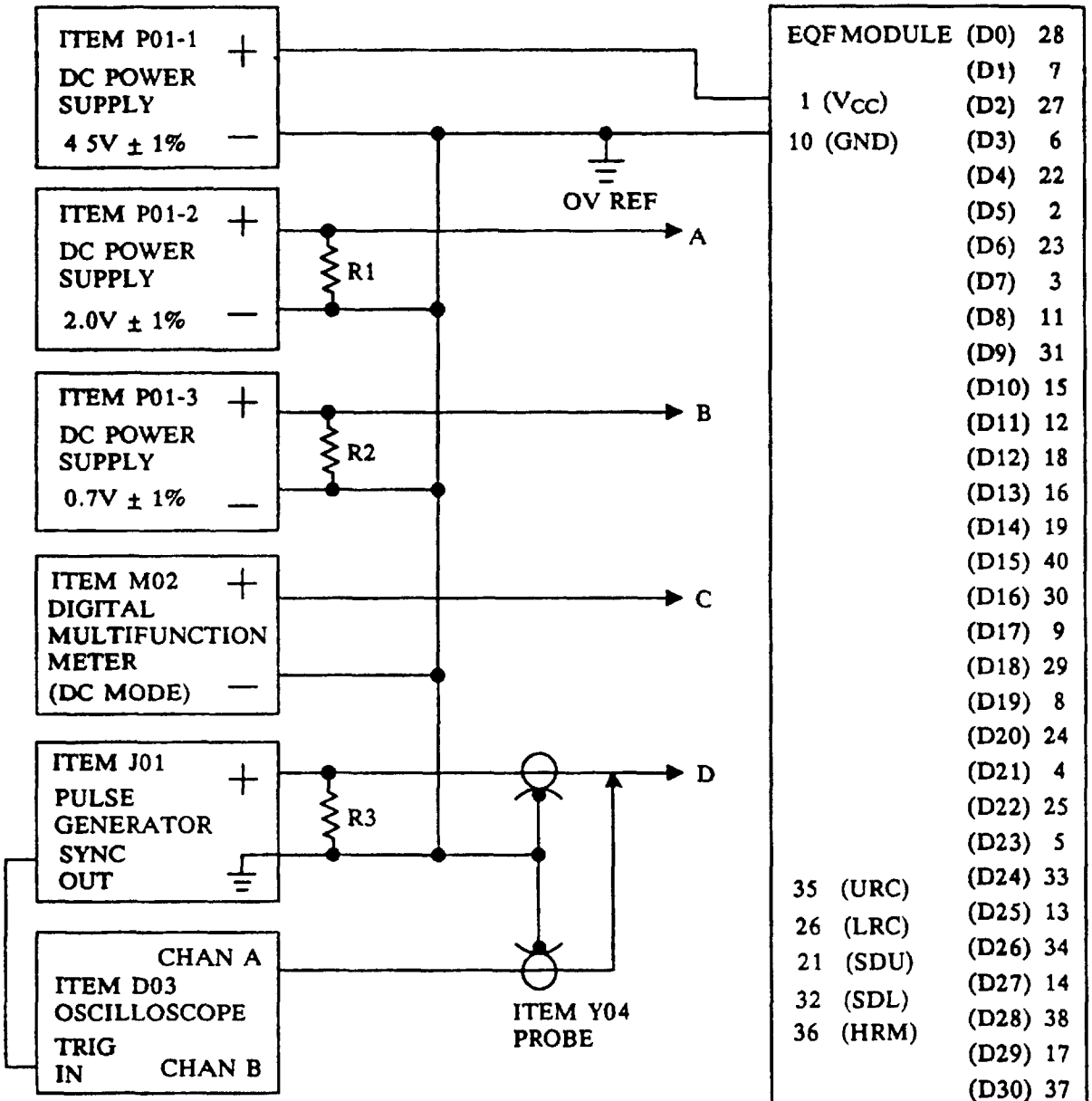
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NOTES:
1. R1 = 1.58KΩ ± 1%, 1/8 W.
2. R2 = 51Ω ± 5%, 1/2W.
3. A minimum of 33 clock pulses are required before point A is connected.
4. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.
5. If tests are successively repeated, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating

FIGURE 7. V_{out} (low-loaded) test circuit for table IV.

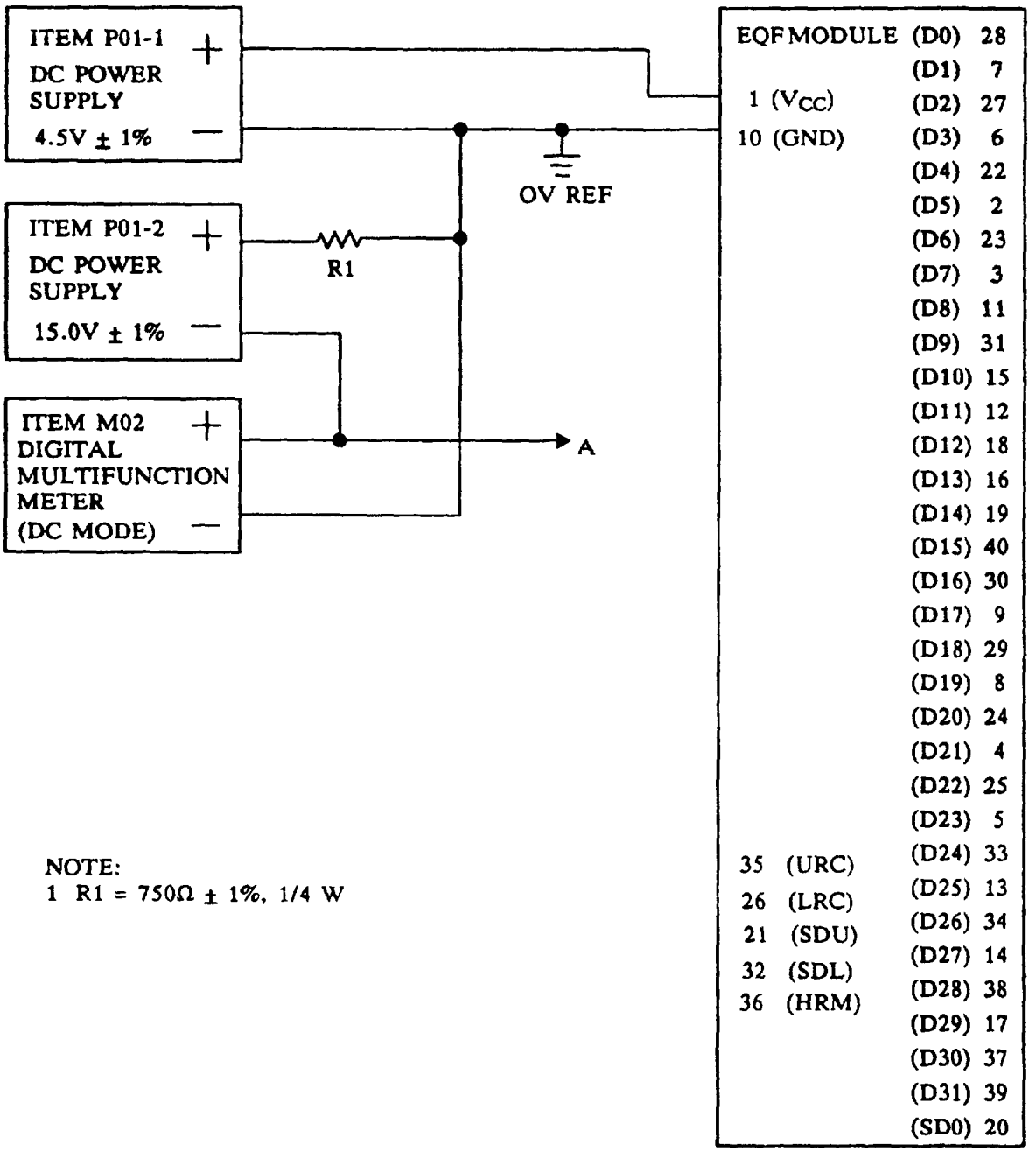
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NOTES:
1 R1 = R2 = 51Ω ± 5%, 1/2W
2. R3 = 51.1Ω ± 1%, 1/4W.

FIGURE 14 Functional test circuit for table VIII.

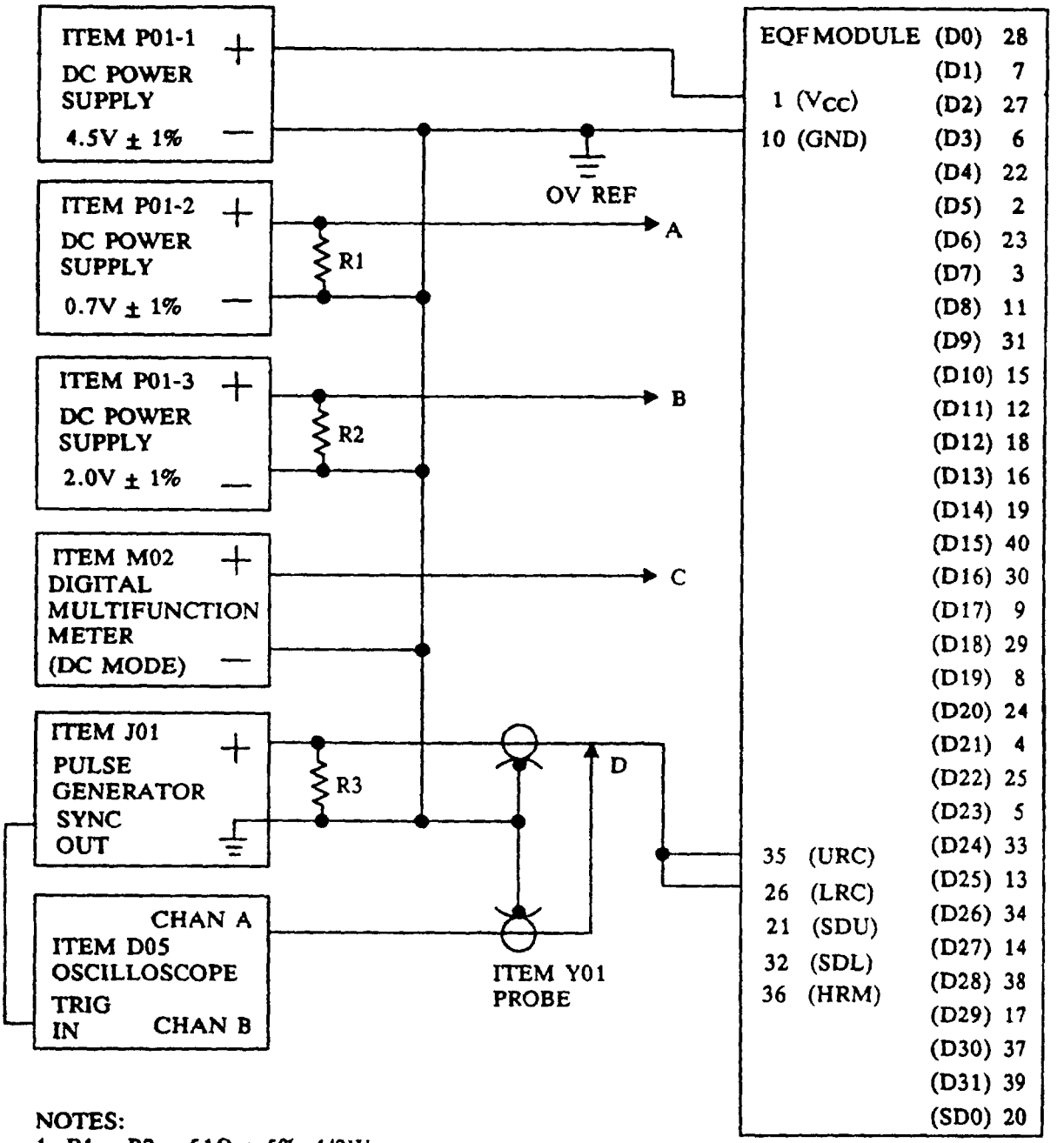
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NOTE:
1 R1 = 750Ω ± 1%, 1/4 W

FIGURE 16. Input clamp diode voltage test circuit for table IX.

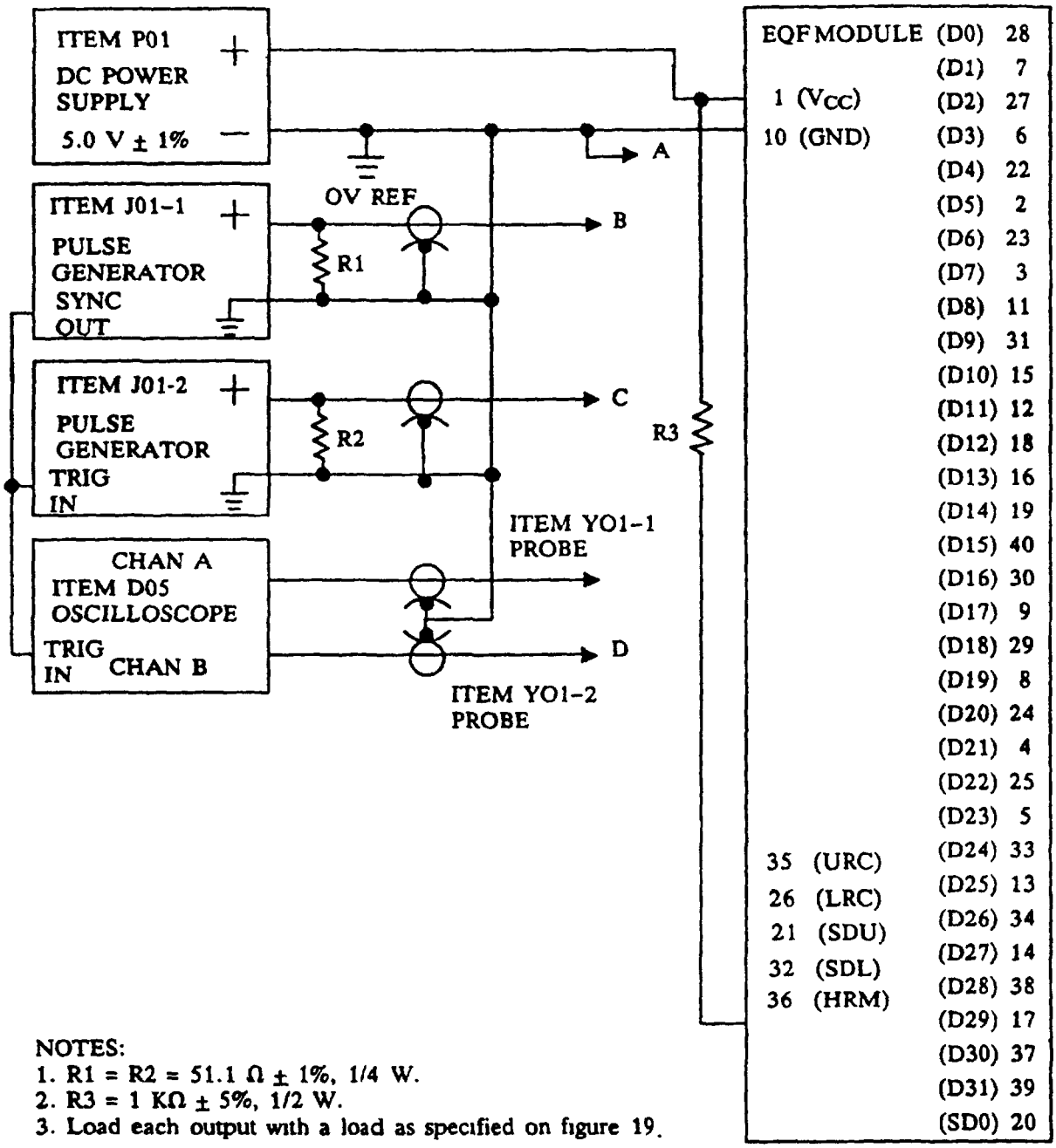
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NOTES:
1. R1 = R2 = 51Ω ± 5%, 1/2W
2. R3 = 51.1Ω ± 1%, 1/4W
3. Load each output with a load as specified on figure 19
4. Allow a minimum of 33 clock pulses before making measurements.

FIGURE 17 Static output voltage (low-level and high-level) test circuit for tables X and XI.

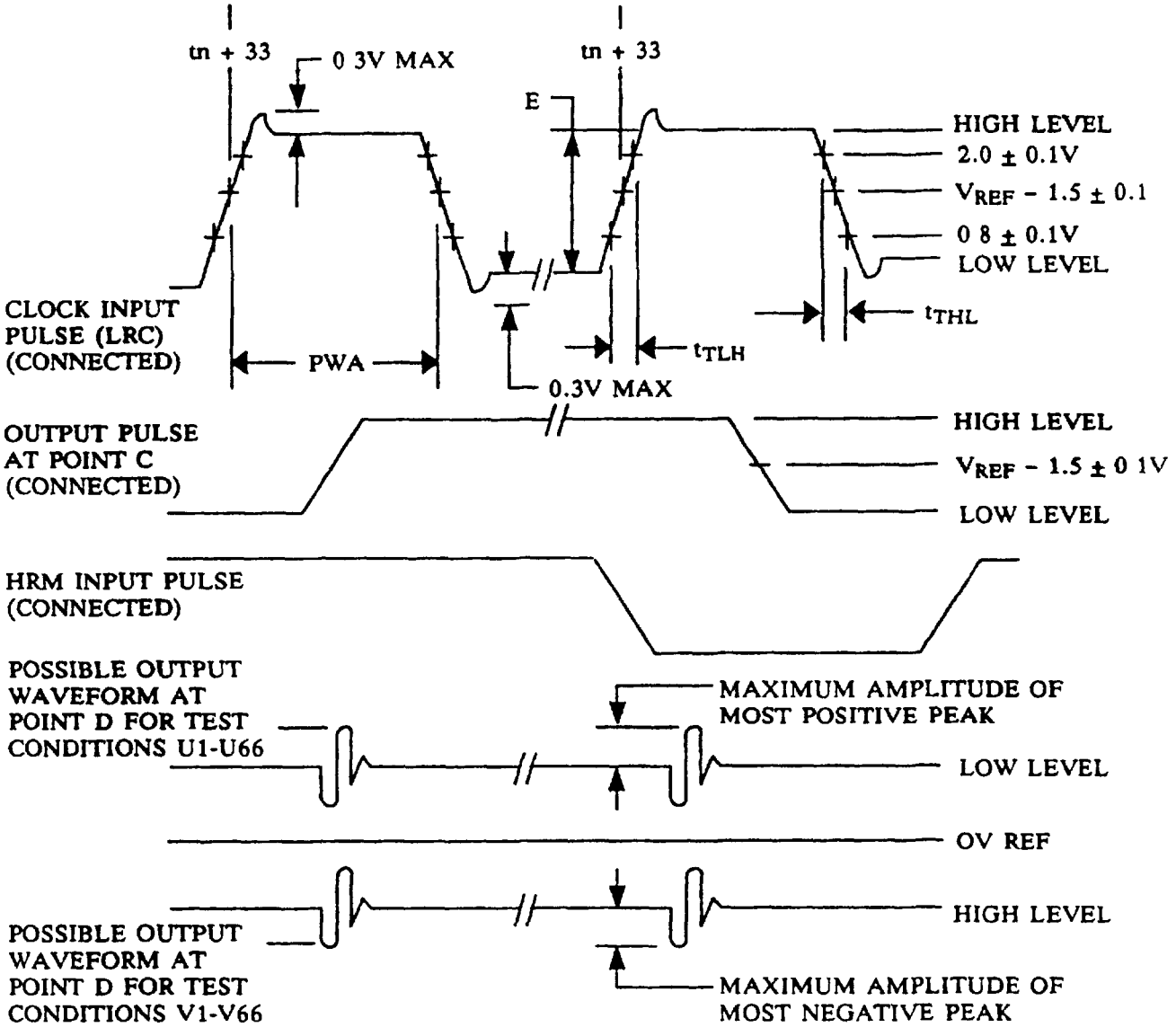
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NOTES:
1. R1 = R2 = 51.1 Ω ± 1%, 1/4 W.
2. R3 = 1 KΩ ± 5%, 1/2 W.
3. Load each output with a load as specified on figure 19.

FIGURE 30 D31 maximum operating frequency check, minimum LRC clock pulse width check, and SDL input data setup and hold time check test circuit for test conditions T5 -T8, table XX.

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NOTES:
1. LRC clock input pulse parameters:
   PWA = 100 ± 10ns.
   $t_{TLH} = t_{THL}$ = 4ns maximum
2. The physical length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25 4mm).
3. All voltages are referenced to 0 V reference unless otherwise specified

FIGURE 38 Output crosstalk (low-level and high-level) test circuit waveforms for tables XXI and XXII.

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